

SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

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Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Built an end-to-end full-stack application using Java, Spring Boot, MySQL, and React, developing 5+ REST APIs and handling 5K+ simulated requests/day.
  - Optimized backend performance through SQL query tuning and indexing, reducing response time by 25% during testing.
  - Implemented Explainable AI-based insights in the React frontend, improving interpretability by 20% and enhancing user experience.

- **All India Council for Technical Education (AICTE)** *July 2024 – September 2024*
  Google AI/ML Virtual Intern Remote
  - Built and evaluated 8+ machine learning models (supervised & unsupervised), achieving up to 92% accuracy across 5+ datasets.
  - Completed 10+ hands-on labs on data preprocessing, feature engineering, and hyperparameter tuning, improving model performance by 25%.
  - Analyzed 6+ real-world AI use cases and implemented 3 bias mitigation techniques to enhance model fairness and reliability.

Projects

- **LLM Instruction Fine-Tuning using LoRA** *Jul 2026*
  Technologies: Python, PyTorch, Hugging Face Transformers, PEFT, LoRA, Llama 3
  - Fine-tuned **Llama 3** using LoRA and Hugging Face PEFT for instruction-following tasks.
  - Built an end-to-end supervised fine-tuning pipeline including preprocessing, training, and inference.
  - Evaluated model performance using BLEU, ROUGE, and qualitative response analysis.

- **Vision Language Model for Document Intelligence** *Mar 2026*
  Technologies: Python, Qwen2-VL, Hugging Face, FastAPI, Streamlit
  - Built a Vision Language Model for document question answering using multimodal transformers.
  - Developed an OCR-free pipeline supporting invoices, IDs, and structured documents.
  - Benchmarked accuracy, F1-score, and inference latency across diverse document layouts.

- **Multimodal Hate Speech Detection using Explainable AI** *Jul 2025*
  Technologies: Python, PyTorch, Hugging Face Transformers, CLIP, BERT, Grad-CAM
  - Developed a multimodal hate speech detection framework using text, image, and metadata fusion.
  - Integrated Explainable AI with Grad-CAM, attention visualization, and token attribution.
  - Conducted ablation studies and evaluated models using Accuracy, F1-score, and ROC-AUC.

- **Marine Sentinel: Real-Time Oil Spill Detection System** *Mar 2026*
  Technologies: YOLOv6, PyTorch, FastAPI, Streamlit, OpenCV
  - Developed a YOLOv6-based oil spill detection model achieving **97% mAP@50** and **86% mAP@50–95**.
  - Built a FastAPI and Streamlit pipeline for real-time UAV video inference.
  - Processed **5,300**+ frames with automated spill localization and severity analysis.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Designed an end-to-end deep learning pipeline on a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Achieved **96%+ classification accuracy** and **AUC up to 0.997** using Neural Networks with stratified 10-fold cross-validation.
  - Reduced feature dimensionality by **75%** using PCA while retaining **95% variance**, improving training efficiency.
  - Improved model performance by **+7% over baseline methods** and enhanced MCC by **+8%** using data augmentation techniques.

Achievements

- Selected as a **Software Engineering Fellow** at Headstarter AI, collaborating on real-world software engineering projects.
- Secured **3rd Place** in CodeMavens 2025 Hackathon organized by ACM-W SRM Student Chapter.
- Built and deployed AI applications across Computer Vision, NLP, Explainable AI, and Full-Stack Software Engineering.

Technical Skills

- **Languages**: Python, Java, C++, C, JavaScript, SQL
- **Developer Tools**: Git, GitHub, Linux, VS Code, Eclipse, Jupyter Notebook, Docker, Postman
- **Frameworks / Libraries**: PyTorch, TensorFlow, Hugging Face Transformers, PEFT, Scikit-learn, Pandas, NumPy, OpenCV, FastAPI, Streamlit, React.js, Node.js
- **LLM / Generative AI**: Large Language Models (LLMs), Vision Language Models (VLMs), Transformers, LoRA, PEFT, LangChain, Hugging Face, Llama 3, Qwen2-VL, BERT, RoBERTa, CLIP, Retrieval-Augmented Generation (RAG)
- **Computer Vision**: YOLOv6, Image Classification, Object Detection, Instance Segmentation, OCR, Explainable AI (Grad-CAM)
- **Research & Evaluation**: Model Fine-tuning, Hyperparameter Tuning, Ablation Studies, Error Analysis, Benchmarking, Model Evaluation, Prompt Engineering
- **Databases**: MySQL, MongoDB
- **Core Computer Science**: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks